

La Union hace la Fuerza



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1 Data Structures

1.1 Priority Queue de minimos

```

1 priority_queue<int, vector<int>, greater<int>> pq;
2 pq.push(x); // Inserta el elemento x
3 pq.top(); // Devuelve el menor elemento (minimo)
4 pq.pop(); // Elimina el menor elemento
5 pq.empty(); // Retorna true si esta vacia
6 pq.size(); // Numero de elementos

```

1.2 Priority Queue de maximos

```

1 priority_queue<int> pq;
2 pq.push(x); // Inserta el elemento x
3 pq.top(); // Devuelve el mayor elemento (maximo)
4 pq.pop(); // Elimina el mayor elemento
5 pq.empty(); // Retorna true si esta vacia
6 pq.size(); // Numero de elementos
7 //Coordenadas cilindricas entra fija

```

1.3 Tabla Aditiva - Partial Sum o Prefix Sum

```

1 vector<int> v = {1, 2, 3, 4, 5};
2 vector<int> prefix(v.size());
3 partial_sum(v.begin(), v.end(), prefix.begin()) //posicion desde la que se
4 //construira);
5 for (int x : prefix)
6     cout << x << " ";
7 // Output: 1 3 6 10 15

```

2 Operaciones con bits en C++

2.1 Tabla bitwise

AND &		
0	0	0
0	1	0
1	0	0
1	1	1

XOR ^		
0	0	0
0	1	1
1	0	1
1	1	0

OR		
0	0	0
0	1	1
1	0	1
1	1	1

NOT ~	
0	1
1	0

2.2 Funciones utilices de C++

2.2.1 popcount

Para obtener el numero de bits encendidos en un entero

```

1 | cout << __builtin_popcount(n) << '\n'; // siendo n un int

```

Version para numeros de 64bits (long long)

```

1 | cout << __builtin_popcountll(n) << '\n'; // siendo n un long long

```

Hack bits

```

1 /*
2  * Bit Hacks

```

```

3  * Useful techniques for working with bit masks.
4  * Widely used in competitive programming for set problems, DP with
   * bitmasks, combinatorics, etc.
5  */
6
7  #include <bits/stdc++.h>
8  using namespace std;
9
10 using ull = unsigned long long;
11
12 /*
13  * next_bits_permutation(x)
14  * Returns the next mask with the same number of set bits (1s) in
   * lexicographical order.
15  * Useful for generating all combinations of k elements as bit masks.
16  *
17  * Example:
18  * x = 0b00111 (represents a combination of 3 elements)
19  * next_bits_permutation(x) -> 0b01011, then 0b01101, etc.
20  */
21 ull next_bits_permutation(ull x) {
22     ull c = __builtin_ctzll(x);          // Count trailing zeros in x
23     ull r = x + (1ULL << c);             // Add 1 to the least
   * significant set bit
24     return (r ^ x) >> (c + 2) | r;       // Calculate the next
   * permutation
25 }
26
27 /*
28  * subsets(s)
29  * Iterates over all proper subsets (does not include s) of a given set
   * s.
30  * The set s is represented as a bit mask.
31  *
32  * Example:
33  * s = 0b1011 (set {0,1,3})
34  * Will iterate over subsets: 1010, 1001, 1000, 0011, 0010, 0001
35  */
36 void subsets(ull s) {
37     for (ull x = s; x;) {
38         --x &= s;
39         // You can work with each subset x here
40         // For example, print it:

```

```

41         // cout << bitset<4>(x) << '\n';
42     }
43 }

```

2.3 Gcd Bits

```

1  //gcd using binary algorithm
2  int binary_gcd(int a, int b) {
3      if (a == 0) return b;
4      if (b == 0) return a;
5
6      int shift;
7      for (shift = 0; ((a | b) & 1) == 0; ++shift) {
8          a >>= 1;
9          b >>= 1;
10     }
11
12     while ((a & 1) == 0)
13         a >>= 1;
14
15     do {
16         while ((b & 1) == 0)
17             b >>= 1;
18
19         if (a > b) swap(a, b);
20         b = b - a;
21     } while (b != 0);
22
23     return a << shift;
24 }

```

3 Grafos

3.1 Revision 8 adyacentes

```

1  vector <int> dx{-1,1,-1,1,0,0,1,-1};
2  vector <int> dy{-1,1,0,0,1,-1,-1,1};

```

3.2 Algoritmo Dijkstra

```

1  /*
2  * Dijkstra's Algorithm

```

```

3  * Finds the minimum distances from a source node in a weighted graph
   * with no negative weights.
4  * Complexity:  $O((V + E) \log V)$ 
   */
6
7  #include <bits/stdc++.h>
8  using namespace std;
9
10 const int INF = 1e9;
11 using pii = pair<int, int>; // {dist, node}
12
13 vector<int> dijkstra(int n, int src, const vector<vector<pii>>& adj) {
14     vector<int> dist(n, INF);
15     priority_queue<pii, vector<pii>, greater<>> pq;
16     dist[src] = 0;
17     pq.emplace(0, src);
18
19     while (!pq.empty()) {
20         auto [d, u] = pq.top(); pq.pop();
21         if (d > dist[u]) continue;
22         for (auto [v, w] : adj[u]) {
23             if (dist[v] > dist[u] + w) {
24                 dist[v] = dist[u] + w;
25                 pq.emplace(dist[v], v);
26             }
27         }
28     }
29     return dist;
30 }

```

3.3 Algoritmo Bellman-Ford

```

1  /*
2  * Bellman-Ford Algorithm
3  * Calculates the shortest distance from a source node, detects negative
   * cycles.
4  * Complexity:  $O(V * E)$ 
   */
6
7  #include <bits/stdc++.h>
8  using namespace std;
9
10 const int INF = 1e9;

```

```

11
12 struct Edge {
13     int u, v, w;
14 };
15
16 vector<int> bellman_ford(int n, int src, const vector<Edge>& edges) {
17     vector<int> dist(n, INF);
18     dist[src] = 0;
19
20     for (int i = 1; i < n; ++i) {
21         for (auto [u, v, w] : edges) {
22             if (dist[u] < INF && dist[v] > dist[u] + w)
23                 dist[v] = dist[u] + w;
24         }
25     }
26
27     // Negative cycle detection
28     for (auto [u, v, w] : edges) {
29         if (dist[u] < INF && dist[v] > dist[u] + w)
30             throw runtime_error("Negative_weight_cycle_detected");
31     }
32
33     return dist;
34 }

```

3.4 Conteo de componentes conexas

```

1  /*
2  * Conteo de componentes conexas en grafo no dirigido
3  */
4
5  #include <bits/stdc++.h>
6  using namespace std;
7
8  int count_components(int n, const vector<vector<int>>& adj) {
9     vector<bool> visited(n, false);
10     int count = 0;
11     for (int u = 0; u < n; ++u) {
12         if (!visited[u]) {
13             dfs(u, adj, visited);
14             ++count;
15         }
16     }

```

```

17     return count;
18 }

```

3.5 Union Find

```

1  /*
2   * Union-Find (Disjoint Set Union - DSU)
3   * Efficient operations for union and find on disjoint sets.
4   * Supports path compression and union by size or rank.
5   * Amortized complexity: O(alpha(n)) per operation
6   */
7
8  #include <bits/stdc++.h>
9  using namespace std;
10
11 struct DSU {
12     vector<int> parent, size;
13
14     DSU(int n) {
15         parent.resize(n);
16         size.assign(n, 1);
17         iota(parent.begin(), parent.end(), 0); // parent[i] = i
18     }
19
20     int find(int u) {
21         if (parent[u] != u)
22             parent[u] = find(parent[u]); // path compression
23         return parent[u];
24     }
25
26     bool unite(int u, int v) {
27         u = find(u), v = find(v);
28         if (u == v) return false;
29         if (size[u] < size[v]) swap(u, v); // union by size
30         parent[v] = u;
31         size[u] += size[v];
32         return true;
33     }
34
35     bool same(int u, int v) {
36         return find(u) == find(v);
37     }
38 };

```

3.6 Graph Diameter

```

1  // Returns the diameter of an undirected unweighted graph
2  // n: number of nodes (1-based), edges: list of {u, v}
3  int graph_diameter(int n, const vector<pair<int, int>>& edges) {
4      const int INF = 1e9;
5      vector<vector<int>> dist(n + 1, vector<int>(n + 1, INF));
6      for (int i = 1; i <= n; i++) dist[i][i] = 0;
7      for (auto [u, v] : edges) {
8          dist[u][v] = 1;
9          dist[v][u] = 1;
10     }
11     for (int k = 1; k <= n; k++)
12         for (int i = 1; i <= n; i++)
13             for (int j = 1; j <= n; j++)
14                 dist[i][j] = min(dist[i][j], dist[i][k] + dist[k][j]);
15
16     int ans = 0;
17     for (int i = 1; i <= n; i++)
18         for (int j = 1; j <= n; j++)
19             if (dist[i][j] < INF)
20                 ans = max(ans, dist[i][j]);
21     return ans;
22 }
23
24 // Example usage:
25 // int n = 4;
26 // vector<pair<int,int>> edges = {{1,2},{2,3},{3,4}};
27 // int d = graph_diameter(n, edges); // d ==

```

4 Manejo de Strings

4.1 Sub String

```

1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  int main() {
6      string s = "ProgramacionCompetitiva";
7
8      string sub1 = s.substr(0, 11); // "Programacion"
9      string sub2 = s.substr(11, 11); // "Competitiva"

```

```

10     string sub3 = s.substr(5);        // from index 5 to the end: "
        amacionCompetitiva"
11
12     cout << sub1 << '\n';
13     cout << sub2 << '\n';
14     cout << sub3 << '\n';
15
16     return 0;
17 }

```

4.2 KMP

```

1  /*
2  * KMP - Knuth-Morris-Pratt
3  * Busca todas las ocurrencias del patron en el texto
4  * Complejidad:  $O(n + m)$ , donde  $n$  = tamaño del texto,  $m$  = tamaño del
        patron
5  */
6
7  vector<int> prefix_function(const string &pat) {
8      int m = pat.size();
9      vector<int> pi(m);
10     for (int i = 1; i < m; ++i) {
11         int j = pi[i - 1];
12         while (j > 0 && pat[i] != pat[j])
13             j = pi[j - 1];
14         if (pat[i] == pat[j])
15             ++j;
16         pi[i] = j;
17     }
18     return pi;
19 }
20
21 vector<int> kmp_search(const string &text, const string &pat) {
22     vector<int> pi = prefix_function(pat);
23     vector<int> occ;
24     int n = text.size(), m = pat.size(), j = 0;
25     for (int i = 0; i < n; ++i) {
26         while (j > 0 && text[i] != pat[j])
27             j = pi[j - 1];
28         if (text[i] == pat[j])
29             ++j;
30         if (j == m) {

```

```

31             occ.push_back(i - m + 1); // ocurrencia encontrada
32             j = pi[j - 1];
33         }
34     }
35     return occ;
36 }
37
38 /*
39 * Z-Algorithm
40 *  $Z[i]$  = longitud del prefijo comun entre  $s$  y  $s[i..]$ 
41 * Sirve para buscar un patron en texto en  $O(n + m)$ 
42 */
43
44 vector<int> z_function(const string &s) {
45     int n = s.size();
46     vector<int> z(n);
47     for (int i = 1, l = 0, r = 0; i < n; ++i) {
48         if (i <= r)
49             z[i] = min(r - i + 1, z[i - l]);
50         while (i + z[i] < n && s[z[i]] == s[i + z[i]])
51             ++z[i];
52         if (i + z[i] - 1 > r)
53             l = i, r = i + z[i] - 1;
54     }
55     return z;
56 }
57
58 // Buscar patron en texto usando Z-function
59 vector<int> z_search(const string &text, const string &pat) {
60     string s = pat + '#' + text;
61     vector<int> z = z_function(s);
62     vector<int> occ;
63     int m = pat.size();
64     for (int i = m + 1; i < (int)s.size(); ++i) {
65         if (z[i] == m)
66             occ.push_back(i - m - 1); // ocurrencia en texto
67     }
68     return occ;
69 }

```

5 Trucos Varios

5.1 Next Permutation

```

1 //Genera todas las permutaciones en Orden lexicografico
2 // La secuencia debe estar previamente ordenada para que funcione
3 vector<int> v = {1, 2, 3};
4 do {for (int x : v)cout << x << '␣';
5     cout << '\n';
6 } while (next_permutation(v.begin(), v.end()));
7 // Salidas 1 2 3 // 1 3 2 // 2 1 3 // 2 3 1 // 3 1 2 // 3 2 1

```

5.2 Uso de Fixed

```

1 #include <iostream>
2 #include <iomanip> // std::fixed, std::setprecision
3 using namespace std;
4
5 int main() {
6     double pi = 3.1415926535;
7     cout << fixed << setprecision(3) << pi << '\n'; // Output: 3.142
8 }

```

5.3 Equivante fixed con printf

```

1 #include <cstdio>
2
3 int main() {
4     double pi = 3.1415926535;
5     printf("%.3f\n", pi); // Output: 3.142
6 }

```

5.4 iota

Para generar arreglos consecutivos

```

1 #include <numeric>
2 vector<int> v(5); // [0, 0, 0, 0, 0]
3 iota(v.begin(), v.end(), 1); // [1, 2, 3, 4, 5]

```

5.5 Conversion de unidades

```

1 /*
2  * Conversiones de unidades

```

```

3  * Autor: Jairo Flores
4  * Conversiones comunes para longitud, masa, tiempo, temperatura, etc.
5  */
6
7 #include <bits/stdc++.h>
8 using namespace std;
9
10 // Longitud
11 double km_to_m(double km) { return km * 1000.0; }
12 double m_to_km(double m) { return m / 1000.0; }
13 double m_to_cm(double m) { return m * 100.0; }
14 double cm_to_m(double cm) { return cm / 100.0; }
15 double inch_to_cm(double in) { return in * 2.54; }
16 double cm_to_inch(double cm) { return cm / 2.54; }
17 double ft_to_m(double ft) { return ft * 0.3048; }
18 double m_to_ft(double m) { return m / 0.3048; }
19
20 // Masa
21 double kg_to_g(double kg) { return kg * 1000.0; }
22 double g_to_kg(double g) { return g / 1000.0; }
23 double lb_to_kg(double lb) { return lb * 0.45359237; }
24 double kg_to_lb(double kg) { return kg / 0.45359237; }
25
26 // Tiempo
27 double hr_to_min(double hr) { return hr * 60.0; }
28 double min_to_sec(double min) { return min * 60.0; }
29 double sec_to_ms(double sec) { return sec * 1000.0; }
30
31 // Temperatura
32 double celsius_to_fahrenheit(double c) { return (c * 9.0 / 5.0) + 32.0; }
33 double fahrenheit_to_celsius(double f) { return (f - 32.0) * 5.0 / 9.0; }
34 double celsius_to_kelvin(double c) { return c + 273.15; }
35 double kelvin_to_celsius(double k) { return k - 273.15; }
36
37 // Velocidad
38 double kmph_to_mps(double kmph) { return kmph / 3.6; }
39 double mps_to_kmph(double mps) { return mps * 3.6; }
40 double mph_to_kmph(double mph) { return mph * 1.60934; }
41 double kmph_to_mph(double kmph) { return kmph / 1.60934; }
42
43 // Area

```

```

44 double m2_to_km2(double m2) { return m2 / 1e6; }
45 double km2_to_m2(double km2) { return km2 * 1e6; }
46 double acre_to_m2(double acre) { return acre * 4046.85642; }
47 double m2_to_acre(double m2) { return m2 / 4046.85642; }
48
49 // Volumen
50 double l_to_ml(double l) { return l * 1000.0; }
51 double ml_to_l(double ml) { return ml / 1000.0; }
52 double gal_to_l(double gal) { return gal * 3.78541; } // gal (US)
53 double l_to_gal(double l) { return l / 3.78541; }
54
55 // Binario
56 double kb_to_bytes(double kb) { return kb * 1024.0; }
57 double mb_to_bytes(double mb) { return mb * 1024.0 * 1024.0; }
58 double gb_to_bytes(double gb) { return gb * 1024.0 * 1024.0 * 1024.0; }
59 double bytes_to_kb(double b) { return b / 1024.0; }
60 double bytes_to_mb(double b) { return b / (1024.0 * 1024.0); }
61 double bytes_to_gb(double b) { return b / (1024.0 * 1024.0 * 1024.0); }
62
63 // Prueba
64 int main() {
65     cout << fixed << setprecision(2);
66     cout << "5_km_=" << km_to_m(5) << "_m\n";
67     cout << "100_F_=" << fahrenheit_to_celsius(100) << "_C\n";
68     cout << "1_acre_=" << acre_to_m2(1) << "_m^2\n";
69     cout << "1.5_GB_=" << gb_to_bytes(1.5) << "_bytes\n";
70     return 0;
71 }

```

5.6 Salidas varias

```

1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     // \n: salto de linea
6     cout << "Hola\nMundo" << endl;
7
8     // \t: tabulacion horizontal
9     cout << "Nombre\tEdad\nJuan\t20\n";
10
11     // \\: imprime el backslash
12     cout << "Ruta:_C:\Users\Jairo\Documentos\n";

```

```

13
14 // \": imprime comillas dobles
15 cout << "Dijo:_\"Hola_mundo\"_n";
16
17 // \': imprime comillas simples
18 cout << "Letra:_\'A\'_n";
19
20 // \a: beep (no siempre funciona en todas consolas)
21 cout << "\a"; // intento de alerta sonora
22
23 // \b: retroceso (borra el caracter anterior en algunas consolas)
24 cout << "ABC\bD\n"; // imprime ABD (borra la C visualmente)
25
26 // \r: retorno de carro (reinicia al inicio de la linea)
27 cout << "Sobrescribe\rHola\n"; // Holaescribe
28
29 // \f y \v: no tienen mucho efecto visible en terminales modernas
30 cout << "Primera_linea\fSegunda_(form_feed)\n";
31 cout << "Primera_linea\vSegunda_(vertical_tab)\n";
32
33 // \0: caracter nulo, marca fin de cadena en arreglos de char
34 char palabra[] = {'H', 'o', 'l', 'a', '\0'};
35 cout << "Cadena_terminada_por_nulo:_ " << palabra << endl;
36
37 return 0;
38 }

```

5.7 Aritmetica con Strings

```

1 #include <bits/stdc++.h>
2 #include <string>
3 #include <algorithm>
4
5 using namespace std;
6
7 // Sum of strings representing positive integers
8 string sumaStrings(const string &a, const string &b) {
9     string res;
10    int carry = 0;
11    int i = (int)a.size() - 1;
12    int j = (int)b.size() - 1;
13
14    while(i >= 0 || j >= 0 || carry) {

```



```

15     int x = i >= 0 ? a[i] - '0' : 0;
16     int y = j >= 0 ? b[j] - '0' : 0;
17     int suma = x + y + carry;
18     carry = suma / 10;
19     res.push_back((suma % 10) + '0');
20     i--; j--;
21 }
22 reverse(res.begin(), res.end());
23 return res;
24 }
25
26 // Compare if a >= b (both positive and without leading zeros)
27 bool mayorIgual(const string &a, const string &b) {
28     if (a.size() != b.size()) return a.size() > b.size();
29     return a >= b;
30 }
31
32 // Subtract strings a - b (assuming a >= b)
33 string restaStrings(const string &a, const string &b) {
34     if (!mayorIgual(a, b)) return "Error: a < b";
35
36     string res;
37     int i = (int)a.size() - 1;
38     int j = (int)b.size() - 1;
39     int borrow = 0;
40
41     while(i >= 0) {
42         int x = a[i] - '0' - borrow;
43         int y = j >= 0 ? b[j] - '0' : 0;
44
45         if (x < y) {
46             x += 10;
47             borrow = 1;
48         } else {
49             borrow = 0;
50         }
51
52         res.push_back((x - y) + '0');
53         i--; j--;
54     }
55
56     // Remove leading zeros
57     while(res.size() > 1 && res.back() == '0') res.pop_back();

```

```

58     reverse(res.begin(), res.end());
59     return res;
60 }
61
62 // Multiplication of strings
63 string multiplicacionStrings(const string &a, const string &b) {
64     int n = (int)a.size();
65     int m = (int)b.size();
66     string res(n + m, '0');
67
68     for(int i = n - 1; i >= 0; i--) {
69         int carry = 0;
70         int x = a[i] - '0';
71         for(int j = m - 1; j >= 0; j--) {
72             int y = b[j] - '0';
73             int suma = (res[i + j + 1] - '0') + x * y + carry;
74             res[i + j + 1] = (suma % 10) + '0';
75             carry = suma / 10;
76         }
77         res[i] += carry;
78     }
79
80     // Remove leading zeros
81     int pos = 0;
82     while (pos < (int)res.size() - 1 && res[pos] == '0') pos++;
83     return res.substr(pos);
84 }
85
86 // Quick test
87 int main() {
88     string A = "123456789123456789";
89     string B = "987654321987654321";
90
91     cout << "Sum: " << sumaStrings(A, B) << "\n";
92     cout << "Subtraction (B - A): " << restaStrings(B, A) << "\n"; // B
93     > A
94     cout << "Multiplication: " << multiplicacionStrings(A, B) << "\n";
95
96     return 0;
97 }

```

5.8 Aritmetica con Strings segun deepseek

```
1 #include <iostream>
2 #include <string>
3 #include <algorithm>
4 using namespace std;
5
6 // Helper function to remove leading zeros
7 string removeLeadingZeros(string num) {
8     size_t nonZeroIndex = num.find_first_not_of('0');
9     if (nonZeroIndex == string::npos) {
10         return "0";
11     }
12     return num.substr(nonZeroIndex);
13 }
14
15 // Addition of two numbers represented as strings
16 string addStrings(string num1, string num2) {
17     num1 = removeLeadingZeros(num1);
18     num2 = removeLeadingZeros(num2);
19
20     int i = num1.length() - 1;
21     int j = num2.length() - 1;
22     int carry = 0;
23     string result;
24
25     while (i >= 0 || j >= 0 || carry > 0) {
26         int digit1 = (i >= 0) ? num1[i--] - '0' : 0;
27         int digit2 = (j >= 0) ? num2[j--] - '0' : 0;
28         int sum = digit1 + digit2 + carry;
29         carry = sum / 10;
30         result.push_back(sum % 10 + '0');
31     }
32
33     reverse(result.begin(), result.end());
34     return result.empty() ? "0" : result;
35 }
36
37 // Compare two numbers represented as strings (returns 1 if num1 > num2,
38 // -1 if num1 < num2, 0 if equal)
39 int compareStrings(string num1, string num2) {
40     num1 = removeLeadingZeros(num1);
41     num2 = removeLeadingZeros(num2);
42
43     if (num1.length() > num2.length()) return 1;
```

```
43     if (num1.length() < num2.length()) return -1;
44
45     for (int i = 0; i < num1.length(); i++) {
46         if (num1[i] > num2[i]) return 1;
47         if (num1[i] < num2[i]) return -1;
48     }
49
50     return 0;
51 }
52
53 // Subtraction of two numbers represented as strings (num1 - num2,
54 // assumes num1 >= num2)
55 string subtractStrings(string num1, string num2) {
56     num1 = removeLeadingZeros(num1);
57     num2 = removeLeadingZeros(num2);
58
59     int i = num1.length() - 1;
60     int j = num2.length() - 1;
61     int borrow = 0;
62     string result;
63
64     while (i >= 0) {
65         int digit1 = num1[i--] - '0' - borrow;
66         int digit2 = (j >= 0) ? num2[j--] - '0' : 0;
67         borrow = 0;
68
69         if (digit1 < digit2) {
70             digit1 += 10;
71             borrow = 1;
72         }
73
74         result.push_back((digit1 - digit2) + '0');
75     }
76
77     reverse(result.begin(), result.end());
78     return removeLeadingZeros(result);
79 }
80
81 // Multiplication of two numbers represented as strings
82 string multiplyStrings(string num1, string num2) {
83     num1 = removeLeadingZeros(num1);
84     num2 = removeLeadingZeros(num2);
```

```
85     if (num1 == "0" || num2 == "0") return "0";
86
87     int n1 = num1.length();
88     int n2 = num2.length();
89     string result(n1 + n2, '0');
90
91     for (int i = n1 - 1; i >= 0; i--) {
92         for (int j = n2 - 1; j >= 0; j--) {
93             int product = (num1[i] - '0') * (num2[j] - '0') + (result[i
94                 + j + 1] - '0');
95             result[i + j + 1] = product % 10 + '0';
96             result[i + j] += product / 10;
97         }
98     }
99     return removeLeadingZeros(result);
100 }
101
102 // Division of two numbers represented as strings (integer division)
103 string divideStrings(string num1, string num2) {
104     num1 = removeLeadingZeros(num1);
105     num2 = removeLeadingZeros(num2);
106
107     if (num2 == "0") return "Error: Division by zero";
108     if (compareStrings(num1, num2) < 0) return "0";
109
110     string quotient;
111     string current;
112
113     for (int i = 0; i < num1.length(); i++) {
114         current += num1[i];
115         current = removeLeadingZeros(current);
116         if (compareStrings(current, num2) < 0) {
117             quotient += "0";
118             continue;
119         }
120
121         int count = 0;
122         string temp = num2;
123         while (compareStrings(current, temp) >= 0) {
124             temp = addStrings(temp, num2);
125             count++;
126         }
```

```
127
128         quotient += to_string(count);
129         temp = multiplyStrings(num2, to_string(count));
130         current = subtractStrings(current, temp);
131     }
132
133     quotient = removeLeadingZeros(quotient);
134     return quotient.empty() ? "0" : quotient;
135 }
136
137 int main() {
138     string num1, num2;
139     char op;
140
141     cout << "Enter the first number: ";
142     cin >> num1;
143     cout << "Enter the second number: ";
144     cin >> num2;
145     cout << "Enter the operation (+, -, *, /): ";
146     cin >> op;
147
148     string result;
149     switch(op) {
150         case '+':
151             result = addStrings(num1, num2);
152             break;
153         case '-':
154             if (compareStrings(num1, num2) < 0) {
155                 result = "-" + subtractStrings(num2, num1);
156             } else {
157                 result = subtractStrings(num1, num2);
158             }
159             break;
160         case '*':
161             result = multiplyStrings(num1, num2);
162             break;
163         case '/':
164             result = divideStrings(num1, num2);
165             break;
166         default:
167             result = "Invalid operation";
168     }
169 }
```

```

170     cout << "Result:_" << result << endl;
171     return 0;
172 }

```

5.9 Problema M codigo

```

1  #include <bits/stdc++.h>
2  using namespace std;
3  vector<int> tabla(10000);
4  int __find(int x) {
5      if (tabla[x] != x)
6          tabla[x] = __find(tabla[x]);
7      return tabla[x];
8  }
9  void __union(int a, int b) {
10     a = __find(a);
11     b = __find(b);
12     if (a != b)
13         tabla[b] = a;
14 }
15 int main() {
16     int n;
17     scanf("%d", &n);
18     vector<int> dx(n + 10);
19     vector<int> dy(n + 10);
20     vector<int> dr(n + 10);
21     for (int i = 0; i < n; i++) {
22         scanf("%d_%d_%d", &dx[i], &dy[i], &dr[i]);
23         tabla[i] = i;
24     }
25     for (int i = 0; i < n; i++) {
26         for (int j = i + 1; j < n; j++) {
27             int hx = dx[i] - dx[j];
28             int hy = dy[i] - dy[j];
29             double tagoras = abs(sqrt(pow(hx, 2) + pow(hy, 2)));
30             if (tagoras <= dr[i] + dr[j]) {
31                 __union(i, j);
32             }
33         }
34     }
35     int ans = 0;
36     for (int u = 0; u < n; u++) {
37         if (__find(u) == u)

```

```

38         ans++;
39     }
40     printf("%d\n", ans);
41 }

```

5.10 Matriz Viborita

```

1  #include <iostream>
2  #include <vector>
3  using namespace std;
4
5  int main() {
6      int n, m;
7      cout << "Enter the number of rows: ";
8      cin >> n;
9      cout << "Enter the number of columns: ";
10     cin >> m;
11
12     vector<vector<int>> matriz(n, vector<int>(m));
13     int valor = 1;
14
15     for (int i = 0; i < n; i++) {
16         if (i % 2 == 0) {
17             // Left to right
18             for (int j = 0; j < m; j++) {
19                 matriz[i][j] = valor++;
20             }
21         } else {
22             // Right to left
23             for (int j = m - 1; j >= 0; j--) {
24                 matriz[i][j] = valor++;
25             }
26         }
27     }
28
29     // Print the matrix
30     cout << "\nSnake matrix:\n";
31     for (const auto& fila : matriz) {
32         for (int x : fila) {
33             cout << x << " ";
34         }
35         cout << "\n";
36     }

```

```

37
38     return 0;
39 }

```

5.11 Triangulo de Pascal

```

1  #include <iostream>
2  #include <vector>
3  #include <iomanip> // for setw
4  using namespace std;
5
6  // Function to calculate combinations C(n, k)
7  int combinacion(int n, int k) {
8      if (k == 0 || k == n) return 1;
9      int res = 1;
10     for (int i = 1; i <= k; ++i) {
11         res *= n--;
12         res /= i;
13     }
14     return res;
15 }
16
17 int main() {
18     int filas;
19     cout << "Enter the number of rows: ";
20     cin >> filas;
21
22     for (int i = 0; i < filas; ++i) {
23         // Print spaces to center
24         cout << setw((filas - i) * 2);
25         for (int j = 0; j <= i; ++j) {
26             cout << setw(4) << combinacion(i, j);
27         }
28         cout << endl;
29     }
30
31     return 0;
32 }

```

6 Geometry

6.1 Basicas

Distancia entre dos puntos: $d(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ (1)

Producto punto: $\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y$ (2)

Producto cruz (2D): $\mathbf{a} \times \mathbf{b} = a_x b_y - a_y b_x$ (3)

Norma de vector: $\|\mathbf{a}\| = \sqrt{a_x^2 + a_y^2}$ (4)

6.2 Triángulos

Área (1/2 base $\times$ altura): $A = \frac{1}{2} |\mathbf{AB} \times \mathbf{AC}|$ (5)

Herón: $s = \frac{a+b+c}{2}$, $A = \sqrt{s(s-a)(s-b)(s-c)}$ (6)

Circunradio: $R = \frac{abc}{4A}$, **Inradio:** $r = \frac{2A}{a+b+c}$ (7)

Incentro: $I = \frac{aA + bB + cC}{a+b+c}$ (8)

6.3 Polígonos

Área (shoelace): $A = \frac{1}{2} \left| \sum_{i=0}^{n-1} (x_i y_{i+1} - x_{i+1} y_i) \right|$ (9)

Centroide: (10)

$$C_x = \frac{1}{6A} \sum_i (x_i + x_{i+1})(x_i y_{i+1} - x_{i+1} y_i),$$
 (11)

$$C_y = \frac{1}{6A} \sum_i (y_i + y_{i+1})(x_i y_{i+1} - x_{i+1} y_i)$$
 (12)

6.4 Círculos

$$\text{Ecuación: } (x - h)^2 + (y - k)^2 = r^2 \quad (13)$$

$$\text{Intersección círculo-círculo:} \quad (14)$$

$$d = \|O_1 O_2\|, \quad a = \frac{R^2 - r^2 + d^2}{2d}, \quad h = \sqrt{R^2 - a^2} \quad (15)$$

$$\text{Puntos } P_1, P_2 = P_0 \pm h \frac{(O_2 - O_1)^\perp}{d} \quad (16)$$

$$\text{Área intersección:} \quad (17)$$

$$A = R^2 \cos^{-1} \left(\frac{d^2 + R^2 - r^2}{2dR} \right) + r^2 \cos^{-1} \left(\frac{d^2 + r^2 - R^2}{2dr} \right) \quad (18)$$

$$- \frac{1}{2} \sqrt{(-d + R + r)(d + R - r)(d - R + r)(d + R + r)} \quad (19)$$

6.5 Vectores y Operaciones

$$\text{Conversión polar} \rightarrow \text{cartesiano: } x = r \cos \theta, \quad y = r \sin \theta \quad (20)$$

$$\text{Conversión cartesiano} \rightarrow \text{polar: } r = \sqrt{x^2 + y^2}, \quad \theta = \text{atan2}(y, x) \quad (21)$$

$$\text{Rotación: } (x', y') = (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta) \quad (22)$$

6.6 Orientación y Segmentos

$$\text{Orient}(A, B, C) = \text{sign}((B - A) \times (C - A)) \quad (23)$$

$$\text{Distancia punto-línea: } d(P, L) = \frac{|Ax_P + By_P + C|}{\sqrt{A^2 + B^2}} \quad (24)$$

$$\text{Intersección segmentos: } \text{Check orientaciones y proyecciones.} \quad (25)$$

6.7 Formulas Basicas

```

1  /*
2  * Geometria basica: calculo de areas y subareas
3  * Autor: Jairo Flores
4  */
5
6  #include <bits/stdc++.h>
7  using namespace std;
8
9  // Area de un triangulo dado base y altura
10 double area_triangulo(double base, double altura) {

```

```

11     return (base * altura) / 2.0;
12 }
13
14 // Area de un rectangulo dado base y altura
15 double area_rectangulo(double base, double altura) {
16     return base * altura;
17 }
18
19 // Area de un cuadrado dado lado
20 double area_cuadrado(double lado) {
21     return lado * lado;
22 }
23
24 // Area de un circulo dado radio
25 double area_circulo(double radio) {
26     const double PI = acos(-1);
27     return PI * radio * radio;
28 }
29
30 // Area de un trapezio (bases mayor y menor, altura)
31 double area_trapezio(double base_mayor, double base_menor, double altura)
32 ) {
33     return ((base_mayor + base_menor) * altura) / 2.0;
34 }
35
36 // Area de un rombo dado diagonal mayor y menor
37 double area_rombo(double diagonal_mayor, double diagonal_menor) {
38     return (diagonal_mayor * diagonal_menor) / 2.0;
39 }
40
41 // Area de un poligono regular (n lados) con apotema y perimetro
42 double area_poligono_regular(int n_lados, double apotema, double
43 perimetro) {
44     return (perimetro * apotema) / 2.0;
45 }
46
47 // Area de un poligono regular usando solo lado y numero de lados (
48 formula basada en apotema)
49 double area_poligono_regular_lado(int n_lados, double lado) {
50     const double PI = acos(-1);
51     double apotema = lado / (2 * tan(PI / n_lados));
52     double perimetro = n_lados * lado;
53     return (perimetro * apotema) / 2.0;

```

```

51 }
52
53 // Area de sector circular (radio y angulo en grados)
54 double area_sector_circular(double radio, double angulo_grados) {
55     const double PI = acos(-1);
56     return (PI * radio * radio) * (angulo_grados / 360.0);
57 }
58
59 // Area de segmento circular (radio y angulo en grados)
60 double area_segmento_circular(double radio, double angulo_grados) {
61     const double PI = acos(-1);
62     double angulo_radianes = angulo_grados * PI / 180.0;
63     double area_sector = (radio * radio * angulo_radianes) / 2.0;
64     double area_triangulo = (radio * radio * sin(angulo_radianes)) /
65         2.0;
66     return area_sector - area_triangulo;
67 }
68
69 // Perimetro de circulo (circunferencia)
70 double perimetro_circulo(double radio) {
71     const double PI = acos(-1);
72     return 2 * PI * radio;
73 }
74
75 // Perimetro de poligono regular (n lados y lado)
76 double perimetro_poligono_regular(int n_lados, double lado) {
77     return n_lados * lado;
78 }
79
80 // Ejemplo de uso
81 int main() {
82     cout << fixed << setprecision(4);
83     cout << "Area triangulo (base=5, altura=3): " << area_triangulo(5,3)
84         << "\n";
85     cout << "Area circulo (radio=2): " << area_circulo(2) << "\n";
86     cout << "Area trapecio (bases=3,5, altura=4): " << area_trapecio
87         (5,3,4) << "\n";
88     cout << "Area poligono regular (hexagono, lado=2): " <<
89         area_poligono_regular_lado(6, 2) << "\n";
90     cout << "Area sector circular (radio=4, angulo=90): " <<
91         area_sector_circular(4, 90) << "\n";
92     return 0;
93 }

```

6.8 Convex Hull

```

1  /*
2   * Convex Hull - Algoritmo Graham Scan
3   * Autor: Jairo Flores
4   * Entrada: vector de puntos (x, y)
5   * Salida: vector de puntos que forman el hull convexo en orden ccw
6   */
7
8  #include <bits/stdc++.h>
9  using namespace std;
10
11 struct Point {
12     long long x, y;
13     bool operator<(const Point& p) const {
14         return y < p.y || (y == p.y && x < p.x);
15     }
16 };
17
18 // Producto cruzado (vectorial) de AB x AC
19 long long cross(const Point &O, const Point &A, const Point &B) {
20     return (A.x - O.x)*(B.y - O.y) - (A.y - O.y)*(B.x - O.x);
21 }
22
23 // Funcion que calcula el convex hull y devuelve los vertices en orden
24 // ccw
25 vector<Point> convexHull(vector<Point> pts) {
26     int n = (int)pts.size();
27     if (n <= 1) return pts;
28
29     // Paso 1: encontrar punto base (con menor y, si empate menor x)
30     sort(pts.begin(), pts.end());
31     Point base = pts[0];
32
33     // Paso 2: ordenar por angulo polar respecto a base
34     sort(pts.begin() + 1, pts.end(), [&](const Point &a, const Point &b)
35         {
36             long long c = cross(base, a, b);
37             if (c == 0) // si son colineales, el mas cercano va primero
38                 return (a.x - base.x)*(a.x - base.x) + (a.y - base.y)*(a.y -
39                     base.y) <
40                     (b.x - base.x)*(b.x - base.x) + (b.y - base.y)*(b.y -
41                         base.y);

```

```

38     return c > 0;
39 });
40
41 // Paso 3: construir el hull usando stack
42 vector<Point> hull;
43 for (auto &p : pts) {
44     while ((int)hull.size() > 1 && cross(hull[hull.size()-2], hull.
45         back(), p) <= 0)
46         hull.pop_back();
47     hull.push_back(p);
48 }
49 return hull;
50 }
51
52 // Ejemplo de uso
53 int main() {
54     ios::sync_with_stdio(false);
55     cin.tie(nullptr);
56
57     int n; cin >> n;
58     vector<Point> pts(n);
59     for (int i = 0; i < n; i++)
60         cin >> pts[i].x >> pts[i].y;
61
62     vector<Point> hull = convexHull(pts);
63
64     cout << (int)hull.size() << "\n";
65     for (auto &p : hull)
66         cout << p.x << " " << p.y << "\n";
67
68     return 0;
69 }

```

6.9 Trigonometria

```

1  /*
2   * Basic trigonometry for competitive programming
3   * Author: Jairo Flores
4   */
5
6 #include <bits/stdc++.h>
7 using namespace std;

```

```

8
9 const double PI = acos(-1);
10
11 // Trigonometric functions
12 // Angles in radians (use degrees * PI/180 to convert)
13
14 // Convert degrees to radians
15 double deg_to_rad(double deg) {
16     return deg * PI / 180.0;
17 }
18
19 // Convert radians to degrees
20 double rad_to_deg(double rad) {
21     return rad * 180.0 / PI;
22 }
23
24 // Calculate hypotenuse with Pythagoras
25 double hypotenuse(double a, double b) {
26     return sqrt(a*a + b*b);
27 }
28
29 // Law of sines: a/sin(A) = b/sin(B) = c/sin(C)
30 double law_of_sines(double a, double A_rad, double B_rad) {
31     // For example, given side a and angle A, calculates side b given
32     // angle B
33     return a * sin(B_rad) / sin(A_rad);
34 }
35
36 // Law of cosines: c^2 = a^2 + b^2 - 2ab cos(C)
37 double law_of_cosines(double a, double b, double C_rad) {
38     return sqrt(a*a + b*b - 2*a*b*cos(C_rad));
39 }
40
41 // Area of a triangle with two sides and the angle between them: (1/2)*
42 // ab*sin(C)
43 double triangle_area_trig(double a, double b, double C_rad) {
44     return 0.5 * a * b * sin(C_rad);
45 }
46
47 // Angle between vectors (example)
48 double angle_between_vectors(double x1, double y1, double x2, double y2)
49 {
50     // Dot product = |v1||v2|cos(theta)

```



```

48     double dot = x1*x2 + y1*y2;
49     double mag1 = sqrt(x1*x1 + y1*y1);
50     double mag2 = sqrt(x2*x2 + y2*y2);
51     return acos(dot / (mag1*mag2)); // returns radians
52 }
53
54 // Distance between two points (x1,y1) and (x2,y2)
55 double distance(double x1, double y1, double x2, double y2) {
56     return sqrt((x1 - x2)*(x1 - x2) + (y1 - y2)*(y1 - y2));
57 }
58
59 // Example usage:
60 int main() {
61     ios::sync_with_stdio(false);
62     cin.tie(nullptr);
63
64     double A_deg = 30.0, B_deg = 45.0;
65     double a = 5.0; // side a
66     double A = deg_to_rad(A_deg);
67     double B = deg_to_rad(B_deg);
68
69     cout << fixed << setprecision(4);
70     cout << "Side b using law of sines: " << law_of_sines(a, A, B) << "\n";
71
72     double c = law_of_cosines(a, law_of_sines(a, A, B), deg_to_rad(60));
73     cout << "Side c using law of cosines: " << c << "\n";
74
75     double area = triangle_area_trig(a, law_of_sines(a, A, B),
76                                     deg_to_rad(60));
77     cout << "Triangle area with sides and angle: " << area << "\n";
78
79     double angle = angle_between_vectors(1, 0, 0, 1);
80     cout << "Angle between vectors (1,0) and (0,1): " << rad_to_deg(
81         angle) << " degrees\n";
82
83     return 0;
84 }

```

7 Math

7.1 Criterios de Divisibilidad

- 2:** el último dígito es par
- 3:** la sumatoria de los dígitos es divisible por 3
- 4:** las dos últimas cifras son 00 o forman un número divisible entre 4
- 5:** la última cifra es 0 o 5
- 6:** es divisible por 2 y por 3 a la vez
- 7:** la diferencia entre el número sin la cifra de las unidades y el doble de la cifra de las unidades es múltiplo de 7
- 8:** las tres últimas cifras son 000 o forman un número divisible por 8
- 9:** la suma de sus cifras es un número divisible entre 9
- 10:** el último dígito es 0

7.2 Factores de un numero

```

1 | vector<ll> all_factors(ll n) {
2 |     vector<ll> fac = {1};
3 |     while (n > 1) {
4 |         const ll p = mpf[n];
5 |         vector<ll> cur = {1};
6 |         while (n % p == 0) {
7 |             n /= p;
8 |             cur.push_back(cur.back() * p);
9 |         }
10 |         vector<ll> tmp;
11 |         for (auto x : fac)
12 |             for (auto y : cur) tmp.push_back(x * y);
13 |         tmp.swap(fac);
14 |     }
15 |     return fac;
16 | }

```

7.3 Linear Sieve

```

1 | constexpr ll MAXN = 1000000;
2 | bitset<MAXN> is_prime;
3 | vector<ll> primes;
4 | ll mpf[MAXN], phi[MAXN], mu[MAXN];

```

```

5
6 void sieve() {
7     is_prime.set();
8     is_prime[1] = 0;
9     mu[1] = phi[1] = 1;
10    for (ll i = 2; i < MAXN; i++) {
11        if (is_prime[i]) {
12            mpf[i] = i;
13            primes.push_back(i);
14            phi[i] = i - 1;
15            mu[i] = -1;
16        }
17        for (ll p : primes) {
18            if (p > mpf[i] || i * p >= MAXN) break;
19            is_prime[i * p] = 0;
20            mpf[i * p] = p;
21            mu[i * p] = -mu[i];
22            if (i % p == 0)
23                phi[i * p] = phi[i] * p, mu[i * p] = 0;
24            else phi[i * p] = phi[i] * (p - 1);
25        }
26    }

```

7.4 Cribita normal

```

1 constexpr int MAXN = 1000000;
2 bitset<MAXN + 1> is_prime;
3
4 void sieve() {
5     is_prime.set(); // Mark all as potentially prime
6     is_prime[0] = is_prime[1] = 0; // 0 and 1 are not prime
7
8     for (int i = 2; i * i <= MAXN; ++i) {
9         if (is_prime[i]) {
10             for (int j = i * i; j <= MAXN; j += i)
11                 is_prime[j] = 0; // Mark multiples of i as not prime
12         }
13     }
14 }

```

7.5 Criba vasito

```

1 int cr[MAXN]; // -1 if prime, some not trivial divisor if not
2 void init_sieve(){

```

```

3     memset(cr, -1, sizeof(cr));
4     for(i, 2, MAXN) if (cr[i] < 0) for (ll j = 1LL * i * i; j < MAXN; j += i) cr[j] = i;
5 }
6 map<int, int> fact(int n) { // must call init_cribe before
7     map<int, int> r;
8     while (cr[n] >= 0) r[cr[n]]++, n /= cr[n];
9     if (n > 1) r[n]++;
10    return r;
11 }

```

7.6 Contar Divisores

```

1 int count_divisors(int n) {
2     int res = 1;
3     for (int i = 2; i * i <= n; ++i) {
4         int count = 0;
5         while (n % i == 0) {
6             n /= i;
7             count++;
8         }
9         res *= (count + 1);
10    }
11    if (n > 1) res *= 2; // n es primo > sqrt(original n)
12    return res;
13 }

```

7.7 Operaciones con matrices

```

1 #include <iostream>
2 #include <vector>
3 #include <cmath>
4 #include <iomanip>
5
6 using namespace std;
7
8 using Matrix = vector<vector<double>>>;
9
10 // Function to print matrix
11 void printMatrix(const Matrix &mat) {
12     for (const auto &row : mat) {
13         for (double val : row) cout << setw(10) << val << " ";
14         cout << "\n";
15     }
16 }

```

```

17
18 // Swap rows r1 and r2 in the matrix
19 void swapRows(Matrix &mat, int r1, int r2) {
20     std::swap(mat[r1], mat[r2]);
21 }
22
23 // Calculate determinant using Gaussian elimination
24 double determinant(Matrix mat) {
25     int n = (int)mat.size();
26     double det = 1.0;
27
28     for (int i = 0; i < n; i++) {
29         // Find row with nonzero pivot
30         int pivot = i;
31         while (pivot < n && fabs(mat[pivot][i]) < 1e-12) pivot++;
32         if (pivot == n) return 0; // singular matrix
33
34         if (pivot != i) {
35             swapRows(mat, i, pivot);
36             det = -det; // change sign if rows are swapped
37         }
38
39         det *= mat[i][i];
40         if (fabs(mat[i][i]) < 1e-12) return 0;
41
42         // Eliminate below the pivot
43         for (int r = i + 1; r < n; r++) {
44             double factor = mat[r][i] / mat[i][i];
45             for (int c = i; c < n; c++) {
46                 mat[r][c] -= factor * mat[i][c];
47             }
48         }
49     }
50
51     return det;
52 }
53
54 // Calculate inverse using Gauss-Jordan
55 bool inverse(const Matrix &input, Matrix &inv) {
56     int n = (int)input.size();
57     Matrix mat = input;
58     inv.assign(n, vector<double>(n, 0));
59

```

```

60 // Create identity matrix in inv
61 for (int i = 0; i < n; i++) inv[i][i] = 1.0;
62
63 for (int i = 0; i < n; i++) {
64     // Find pivot
65     int pivot = i;
66     for (int r = i + 1; r < n; r++) {
67         if (fabs(mat[r][i]) > fabs(mat[pivot][i])) pivot = r;
68     }
69
70     if (fabs(mat[pivot][i]) < 1e-12) return false; // singular
71     // matrix
72
73     if (pivot != i) {
74         swapRows(mat, i, pivot);
75         swapRows(inv, i, pivot);
76     }
77
78     double pivot_val = mat[i][i];
79     for (int c = 0; c < n; c++) {
80         mat[i][c] /= pivot_val;
81         inv[i][c] /= pivot_val;
82     }
83
84     // Eliminate other elements in column i
85     for (int r = 0; r < n; r++) {
86         if (r != i) {
87             double factor = mat[r][i];
88             for (int c = 0; c < n; c++) {
89                 mat[r][c] -= factor * mat[i][c];
90                 inv[r][c] -= factor * inv[i][c];
91             }
92         }
93     }
94
95     return true;
96 }
97
98 // Example usage
99 int main() {
100     Matrix arr = {
101         {4, 7, 2},

```

```

102     {3, 6, 1},
103     {2, 5, 1}
104 };
105
106 cout << "Original_matrix:\n";
107 printMatrix(arr);
108
109 double det = determinant(arr);
110 cout << "\nDeterminant:_" << det << "\n";
111
112 if (fabs(det) < 1e-12) {
113     cout << "Matrix_has_no_inverse_(det=0)\n";
114 } else {
115     Matrix inv;
116     if (inverse(arr, inv)) {
117         cout << "\nInverse_matrix:\n";
118         printMatrix(inv);
119     } else {
120         cout << "Could_not_compute_the_inverse\n";
121     }
122 }
123
124 return 0;
125 }

```

7.8 Fast Exponentiation

```

1 def fastExponentiation(base, exponent):
2     result = 1
3     base = base % MOD
4     while exponent > 0:
5         if (exponent % 2) == 1:
6             result = (result * base) % MOD
7             exponent = exponent // 2
8             base = (base * base) % MOD
9     return result

```

7.9 Euler's Totient Function

```

1 int phi(int n) {
2     int result = n;
3     for (int i = 2; i * i <= n; i++) {
4         if (n % i == 0) {
5             while (n % i == 0)

```

```

6             n /= i;
7             result -= result / i;
8         }
9     }
10    if (n > 1)
11        result -= result / n;
12    return result;
13 }

```

7.10 Simpson's Integration

```

1 const int N = 1000 * 1000; // number of steps (already multiplied by 2)
2
3 double simpson_integration(double a, double b){
4     double h = (b - a) / N;
5     double s = f(a) + f(b); // a = x_0 and b = x_2n
6     for (int i = 1; i <= N - 1; ++i) { // Refer to final Simpson's
7         // formula
8         double x = a + h * i;
9         s += f(x) * ((i & 1) ? 4 : 2);
10    }
11    s *= h / 3;
12    return s;
13 }

```

8 Mathematical formulas

8.1 Sumatorias

$$\sum_{i=1}^n i = \frac{n(n+1)}{2} \quad (26)$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6} \quad (27)$$

$$\sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2} \right)^2 \quad (28)$$

$$\sum_{i=1}^n i^4 = \frac{n^2(n+1)^2(2n+1)}{30} \quad (29)$$